

# Cloud Services

Training / Learning Documentation

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## Software as a Service (SaaS)

### Simple Definition

**SaaS** is a cloud model where:

You use a software application through the internet, without installing or managing anything.

### What actually happens (behind the scenes)

When you use SaaS:

- The **software** runs on cloud servers
- The **data** is stored in the cloud
- The **updates** are done automatically
- You just **open and use it in a browser**

You don't worry about:

- Installation
- Hardware
- Maintenance
- Updates

### Who does what?

| Task             | Who handles it |
|------------------|----------------|
| Software running | Provider       |
| Servers          | Provider       |
| Storage          | Provider       |
| Updates          | Provider       |
| Security (basic) | Provider       |

|               |     |
|---------------|-----|
| Using the app | YOU |
|---------------|-----|

## Real Example (Very Important)

### Google Docs

#### What you do:

- Open browser
- Go to Google Docs
- Start typing

#### What Google does:

- Saves your document automatically
- Stores it online
- Updates the software
- Handles backups

## Why this is SaaS

Because:

- You didn't install anything
- You didn't manage servers
- You didn't update the software

You simply **used the application as a service**

**SaaS = "Use software directly from the internet"**

## Platform as a Service (PaaS)

### Simple Definition

**PaaS** is a cloud model where:

You get a ready-made platform to build, run, and deploy applications without managing servers or infrastructure.

## First understand “Platform”

A **platform** includes everything needed to run an application:

- Operating System (Windows/Linux)
- Runtime (Java, Python, Node.js)
- Web server
- Database support

Normally, setting all this up takes time and effort. In PaaS, it is already prepared for you.

## What actually happens

- You **write code** (your application)
- Upload it to the platform
- The cloud runs it for you

You don't worry about:

- Server setup
- OS installation
- Configuration

## Who does what?

| Task             | Who handles it |
|------------------|----------------|
| Servers          | Provider       |
| Operating System | Provider       |
| Runtime          | Provider       |
| Application code | YOU            |
| Data             | YOU            |

# Real-world Example

## Azure App Service

### What you do:

- Write your website/app code
- Upload it to Azure

### What Azure does:

- Provides server
- Runs your code
- Handles scaling
- Keeps app online

## Why this is PaaS

Because:

- You are not just using software (like SaaS)
- You are **building your own application**
- But you don't manage infrastructure

**PaaS = “Build and run apps without worrying about infrastructure”**

# Infrastructure as a Service (IaaS)

## Simple Definition

**IaaS** is a cloud model where:

You rent basic computing resources like servers, storage, and networking, and build everything else yourself.

## First understand “Infrastructure”

Infrastructure means the **foundation of IT systems**:

- Servers (computers)
- Storage (hard disk space)
- Networking (internet connections, IPs)

In traditional setup, you physically buy all this.

In IaaS, you **rent it virtually**.

## What actually happens

- You create a **virtual machine (VM)**
- Install your own **Operating System**
- Install software
- Run your application

You are responsible for:

- Setup
- Configuration
- Security (to a large extent)

## Who does what?

| Task             | Who handles it |
|------------------|----------------|
| Physical servers | Provider       |
| Data center      | Provider       |
| Network          | Provider       |
| Operating System | YOU            |
| Applications     | YOU            |
| Data             | YOU            |

# Real Example

## Microsoft Azure Virtual Machines

### What you do:

- Create a VM
- Choose OS (Windows/Linux)
- Install software (like a web server)
- Deploy your app

### What Azure does:

- Provides hardware
- Maintains data center
- Ensures connectivity

## Why this is IaaS

Because:

- You are not just using software (SaaS)
- Not just building apps (PaaS)
- You are **managing the system itself**

**IaaS = “Rent infrastructure and control everything else”**

## Summary

- **SaaS (Software as a Service):** You just use the software through the internet; everything else is managed by the provider.  
Example: Google Docs
- **PaaS (Platform as a Service):** You build and run your applications, while the provider manages the platform and infrastructure.  
Example: Azure App Service
- **IaaS (Infrastructure as a Service):** You get virtual servers and control everything from OS to applications; only hardware is managed by the provider.  
Example: Microsoft Azure Virtual Machines

